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with improvements $\nabla_\mu J^\mu_X$ differing by boundary choices (A4/A5). Closed representatives for J^μ are given on FRW/weak–field backgrounds.

Notation and key relation. We use the preamble shorthands $\Sigma_\epsilon \equiv \Pi_{\rm PT}[(\partial\epsilon)^2]$, $\mathrm{sgn}(\Sigma_\epsilon) \equiv \mathrm{sgn}(\Sigma_\epsilon)$, $\hat n_\mu \equiv \dfrac{\partial_\mu\epsilon}{\sqrt{|\Sigma_\epsilon|}}$, $\mathcal{T}^\mu{}_\nu \equiv \mathrm{sgn}(\Sigma_\epsilon)\Big(\hat n^\mu\hat n_\nu - \frac14\delta^\mu{}_\nu\Big)$, $\tau \equiv \hat n^\mu T_\mu = 3\eta\,\mathrm{sgn}(\Sigma_\epsilon)\sqrt{|\Sigma_\epsilon|}$ so that $\mathrm{Tr}\mathcal{T}=0$, $\mathrm{Tr}(\mathcal{T}^2)=\frac34$, and (after C1)

$$\tau^2 = 9\eta^2|\Sigma_\epsilon| = \frac{3}{2}\,(-\,\mathrm{sgn}(\Sigma_\epsilon) I_T), \qquad \mathrm{sgn}(\Sigma_\epsilon)\, I_T \equiv \mathrm{sgn}(\Sigma_\epsilon) I_T = -6\eta^2|\Sigma_\epsilon|.$$

Appendix C.1. rank-one determinant route: Determinant Algebra with the $\frac23$ Normalization

Consider the rank-one determinant route Lagrangian

$$\mathcal{L}_{\rm ROD} = \sqrt{-g}\Big(\sqrt{\det\big[\mathbf{1}+\tfrac23\lambda\,\tau\,\mathcal{T}\big]}-1\Big), \qquad \lambda\in\mathbb{R}. \tag{A30}$$

Using $\sqrt{\det(\mathbf{1}+X)} = 1+\frac12\mathrm{Tr}X+\frac18[(\mathrm{Tr}X)^2-2\mathrm{Tr}X^2]+\mathcal{O}(X^3)$ and $\mathrm{Tr}\mathcal{T}=0$, the quadratic piece is

$$\begin{aligned}\delta^2\mathcal{L}_{\rm ROD} &= \sqrt{-g}\,\frac18\Big[-2\,\mathrm{Tr}\Big((\tfrac23\lambda\,\tau)^2\mathcal{T}^2\Big)\Big] = -\frac14(\tfrac23)^2\lambda^2\,\sqrt{-g}\,\tau^2\,\mathrm{Tr}(\mathcal{T}^2)\\ &= -\frac14\cdot\frac49\cdot\frac34\,\lambda^2\,\sqrt{-g}\,\tau^2 = -\frac{1}{12}\,\lambda^2\,\sqrt{-g}\,\tau^2.\end{aligned} \tag{A31}$$

With $\tau^2 = 9\eta^2|\Sigma_\epsilon|$ and $\mathrm{sgn}(\Sigma_\epsilon)\, I_T = -6\eta^2|\Sigma_\epsilon|$,

$$-\frac{1}{12}\lambda^2\tau^2 = -\frac{1}{12}\lambda^2\,(9\eta^2|\Sigma_\epsilon|) = -\frac34\lambda^2\eta^2|\Sigma_\epsilon| = \Big(\frac{\lambda^2}{8}\Big)\,\mathrm{sgn}(\Sigma_\epsilon)\, I_T,$$

so that

$$\boxed{\delta^2\mathcal{L}_{\rm ROD} = A_\star\,\sqrt{-g}\,\mathrm{sgn}(\Sigma_\epsilon)\, I_T \;+\; \nabla_\mu J^\mu_{\rm ROD}, \qquad A_\star = \frac{\lambda^2}{8}.} \tag{A32}$$

At the bulk-density level one may take $J^\mu_{\rm ROD}\equiv 0$. For unified boundary diagnostics we adopt a common canonical representative $J^\mu_{\rm can}$ for *all* three routes (Section C.4).

Appendix C.2. Closed–Metric Route: Rank–One Deformation Equals rank-one determinant route to $\mathcal{O}(\mathcal{T}^2)$

Take the rank–one deformation $\tilde g_{\mu\nu} = g_{\mu\nu} + \frac23\lambda\,\tau\,\hat n_\mu\hat n_\nu$ so that $\sqrt{-\tilde g} = \sqrt{-g}\sqrt{\det[\mathbf{1}+\frac23\lambda\,\tau\,\mathcal{T}]}$ and define $\mathcal{L}_{\rm CM} \equiv \sqrt{-\tilde g} - \sqrt{-g}$. The same algebra gives

$$\boxed{\delta^2\mathcal{L}_{\rm CM} = A_\star\,\sqrt{-g}\,\mathrm{sgn}(\Sigma_\epsilon)\, I_T \;+\; \nabla_\mu J^\mu_{\rm CM}, \qquad A_\star = \frac{\lambda^2}{8}.} \tag{A33}$$

Thus rank-one determinant route and CM have the *same bulk* coefficient $A_\star$ and differ only by improvements.

Appendix C.3. PT–Even CS/Nieh–Yan Shadow: Quadratic Reduction (σ_ϵ)

Using $T^A\wedge T_A = d(e^A\wedge T_A) - e^A\wedge e^B\wedge R_{AB}$, applying $*$ and the scalar PT projector (A2), and evaluating after C1, the PT–even piece reduces at quadratic order to

$$\boxed{\delta^2\mathcal{L}_{\rm CS^+} = A_\star\,\sqrt{-g}\,\mathrm{sgn}(\Sigma_\epsilon)\, I_T \;+\; \nabla_\mu J^\mu_{\rm CS^+}, \qquad A_\star = \frac{\lambda^2}{8},} \tag{A34}$$

where the Nieh–Yan assignment (A5) reshuffles only boundary conventions inside the PT–even sector.

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